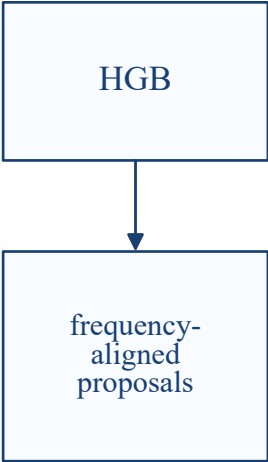


# Dual-mode provenance triage for exploratory replacement proposals

## A. Dual-mode proposal routes

Conservative Mode

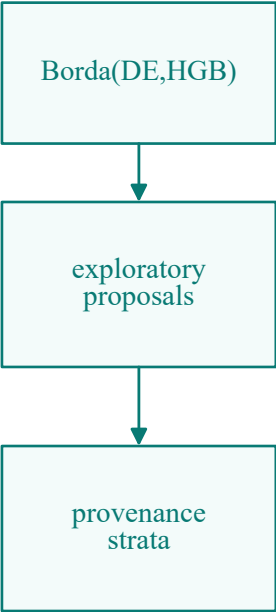
*HGB route*



*Driven by HGB*

Exploration Mode

*Borda(DE,HGB) route*



*Driven by Borda(DE,HGB)*

## B. Provenance strata

G4 shared

$$K_{\text{HGB}} \cap K_{\text{Borda}}$$

consensus

G3 DE-elevated

$$K_{\text{Borda}} \setminus K_{\text{HGB}}$$

similarity-supported  
expansion

G2 Borda-emergent

$$K_{\text{Borda}} \setminus (K_{\text{HGB}} \cup K_{\text{DE}})$$

highest novelty

## C. Coverage-limited A4C triage

| Group | Coverage                          | Alert signal                                  |
|-------|-----------------------------------|-----------------------------------------------|
| G2    | 100%<br>                          | 46.85%                                        |
| G3    | 100%<br>                          | 9.67%                                         |
| G4    | 5.63%<br><br>covered      unknown | 17.60%<br>among covered;<br>94.37%<br>unknown |

A4C = computational triage only;  
not experimental validation.